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Variables Correlated with Retention in High Churn Rate Groups in the Telecommunications Industry

1 Introduction

In the telecommunications (telecom) industry, customer "churn" represents the proportion of customers who terminate their engagement with a service provider. In this competitive market, where average annual churn rates can reach 25%, prioritizing customer retention over acquisition has become imperative due to the significantly higher costs associated with acquiring new customers. Retaining highly profitable customers stands as a primary objective for many providers, emphasizing the need to accurately forecast and manage the churn risk of individual customers.

This study aims to identify the variables that are correlated with customer retention in high churn rate groups within this telecoms company. The findings will help inform what proactive measures might improve customer retention and reduce churn. In this paper, I present the data analysis and findings, along with recommendations for action and suggestions for further analysis.

This project uses data and summary information from Western Governor's University's Master's of Data Analytics program. It is coded in python and uses the following packages: pandas, NumPy, Matplotlib, scikit-learn, Seaborn, SciPy, and IPython.

2 Data Preparation

To prepare the data for analysis, certain variables such as customer ID, survey responses, and miscellaneous information were excluded from the initial data frame. These variables were less informative and were removed to improve the analysis results. A table containing the 30 variables that were included for the analysis is provided below and a detailed description of each variable can be found in the accompanying data dictionary file. Data preprocessing involved standardizing quantitative variables, converting yes/no values to ones and zeros, and applying

one-hot encoding to categorical variables. The final data frames consisted of 10,000 rows of data.

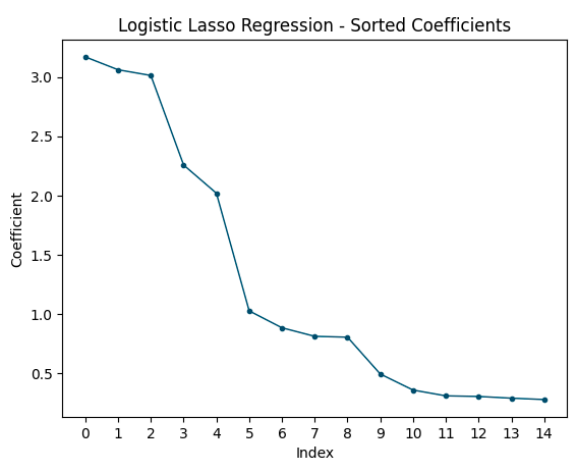
Variable	Data Type	Description
State	Categorical	State of residence
Area	Categorical	Area type (rural, urban, suburban)
Children	Quantitative	Number of children
Age	Quantitative	Age in years
Income	Quantitative	Income in US dollars
Marital	Categorical	Marital status
Gender	Categorical	Gender (male, female, unknown)
Churn	Categorical	Whether customer discontinued services
Outage_sec_perweek	Quantitative	System outages per week in seconds
Email	Quantitative	Number of emails sent to customer in last year
Contacts	Quantitative	Number of times customer contacted support
Yearly_equip_failure	Quantitative	Yearly equipment failures
Techie	Categorical	Whether customer considers themselves “techie”
Contract	Categorical	Time frame of contract
Port_modem	Categorical	Whether customer owns portable modem
Tablet	Categorical	Whether customer owns a tablet
Internet Service	Categorical	Internet service type
Phone	Categorical	Whether customer has a phone service
Multiple	Categorical	Whether customer has multiple lines
OnlineSecurity	Categorical	Whether customer got the security add-on
OnlineBackup	Categorical	Whether customer got the backup add-on
DeviceProtection	Categorical	Whether customer got the protection add-on
TechSupport	Categorical	Whether customer got the tech support add-on
StreamingTV	Categorical	Whether customer has TV streaming
StreamingMovies	Categorical	Whether customer has movie streaming
PaperlessBilling	Categorical	Whether customer uses paperless billing
PaymentMethod	Categorical	Type of payment method
Tenure	Quantitative	Number of months since the start of first contract
MonthlyCharge	Quantitative	Monthly charge for services
Bandwidth_GB_Year	Quantitative	Total bandwidth usage per year

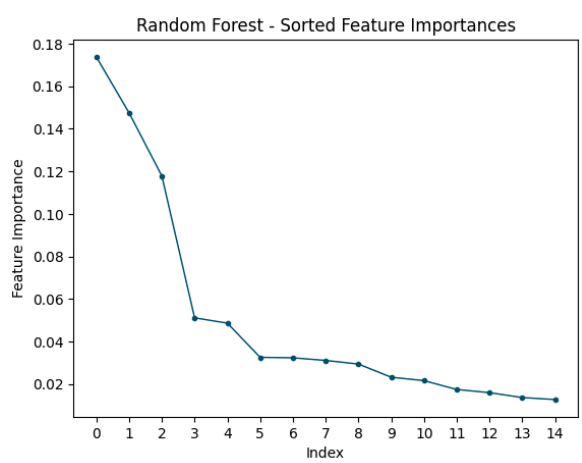
3 Factors Correlated with High Churn Rates

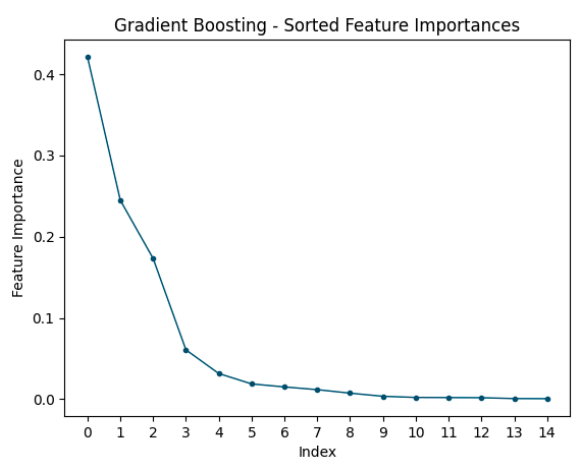
3.1 Identification

The first portion of my analysis focused on determining the factors that have significant correlation with high churn rates. Three feature selection techniques were employed to identify these significant variables: logistic lasso regression, random forest classification, and gradient boosting. Instead of employing the typical train-test splits used with these models, I opted to split the data solely into dependent (churn) and independent variables for feature selection. Since this part of the analysis only aims to describe correlation rather than predict outcomes, I found it unnecessary to include a train-test split.

The models were fit to the data, and the coefficients from logistic lasso regression and feature importance values from random forest and gradient boosting classifiers were extracted and sorted. The top fifteen coefficients/feature importance values for each method were visually examined similar to a scree plot; these plots can be seen below. The indices of the sorted variables were used as the x-axis and allowed me to place the most significant groupings of variables into a list.



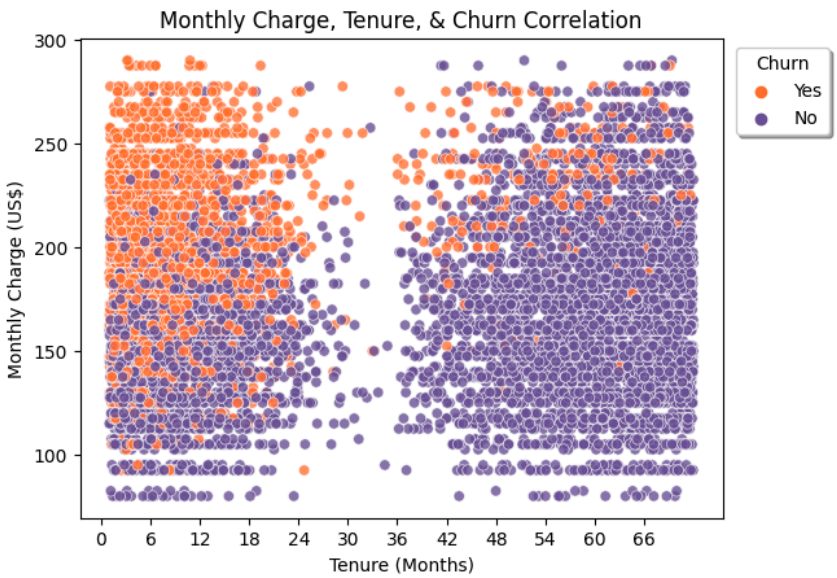




There were five variables selected from the logistic lasso regression, three variables selected from the random forest classification, and three variables selected from the gradient boosting. There were only two variables that appeared significant in all three models: monthly charge and tenure. These variables were selected for further analysis.

3.2 Correlation Visualization

To better understand the relationship between churn, monthly charge, and tenure, I created a scatter plot using monthly charge as the y-axis, tenure as the x-axis, and coloring the data points using churn.

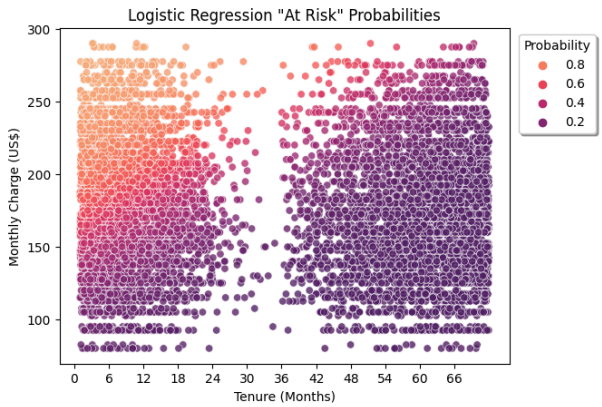


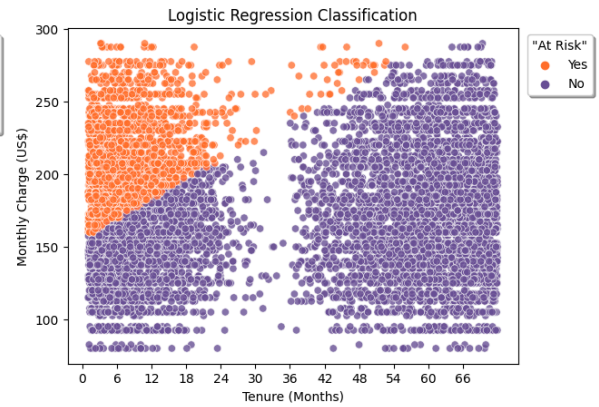
The scatterplot reveals a clear grouping of high churn rates in the high monthly charge, short tenure region of data points. The region least associated with high churn rates is the low monthly charge, long tenure region of data points.

3.3 Defining “At Risk” Customers

Building upon the previous analysis, I decided to focus on identifying the variables most associated with customer retention in the high churn rate group. To achieve this, I defined a subset of customers as the "At Risk" group based on a sufficiently high monthly charge and sufficiently short tenure. Logistic regression was used to determine the cutoff point for this group. Both the probability and the prediction values were stored as separate variables in the data

frame. I then used these variables to visually confirm that the logistic regression captured the high churn rate regions of the data well.





4. Variables Correlated with Retention in “At Risk” Subset

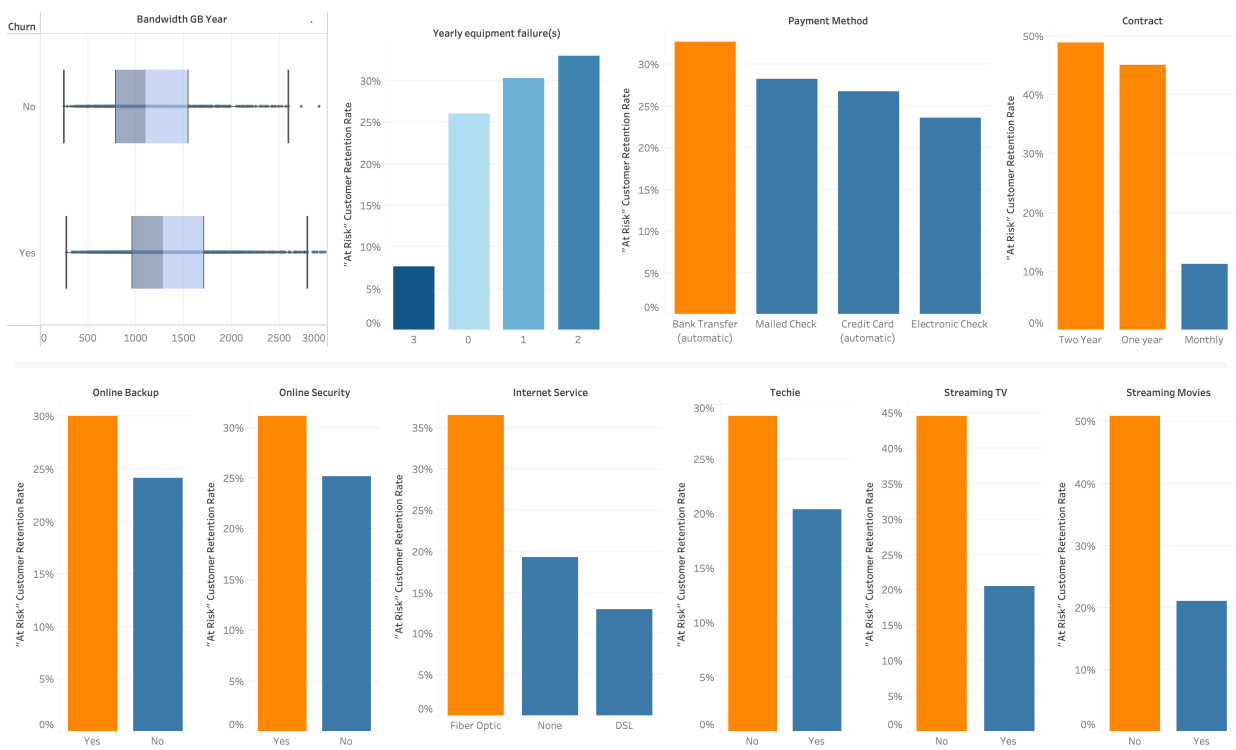
Once the "At Risk" subset was established, I compared the variables between customers who discontinued their services and those who were predicted to do so, but remained customers. Many of the quantitative variables in this dataset are non-normal or ordinary measured, so for the quantitative variables, I employed a Mann-Whitney U test, while a chi-square test was used for categorical variables. With a significance level set to 0.05, the following variables were identified as statistically significant:

Variable	Data Type	Description
Yearly_equip_failure	Quantitative	Yearly equipment failures
Bandwidth_GB_Year	Quantitative	Total bandwidth usage per year
Techie	Categorical	Whether customer considers themselves “techie”
Contract	Categorical	Time frame of contract
Internet Service	Categorical	Internet service type
OnlineSecurity	Categorical	Whether customer got the security add-on
OnlineBackup	Categorical	Whether customer got the backup add-on
StreamingTV	Categorical	Whether customer has TV streaming
StreamingMovies	Categorical	Whether customer has movie streaming
PaymentMethod	Categorical	Type of payment method

With a Bonferroni correction to adjust for multiple tests, the following variables were identified as statistically significant:

Variable	Data Type	Description
Bandwidth_GB_Year	Quantitative	Total bandwidth usage per year
Techie	Categorical	Whether customer considers themselves “techie”
Contract	Categorical	Time frame of contract
Internet Service	Categorical	Internet service type
StreamingTV	Categorical	Whether customer has TV streaming
StreamingMovies	Categorical	Whether customer has movie streaming

To gain deeper insights into the relationship between these variables and churn, I created a visualization dashboard using Tableau. I selected the variables identified with a significance level set to 0.05 to get a broader understanding of the trends in the data. The dashboard displays the retention rate as a percentage of "at risk" customers.



5 Conclusion

5.1 Findings

Customers in the "at risk" group who bundled select telecom services into yearly contracts and used automatic bank transfers exhibited higher retention rates. Conversely, customers who subscribed to streaming TV and movie services, considered themselves “techie”, used more bandwidth per year, and had three equipment failures per year showed lower retention rates. All of the variables positively correlated with retention are controllable by the telecoms company. However, all of the variables negatively correlated with retention are mostly out of the control of the telecoms company. That being said, I believe there is actionable insight contained for both types of correlation.

5.2 Recommended Course of Action

Based on these findings, I recommend the following courses of action: First, this telecoms company should consider initiating a promotional campaign that offers bundled services to customers starting a yearly contract. The bundled services should include any of the following: online backup, online security, and fiber optic internet. Second, providing a one-time incentive for customers to set up automatic bank transfers. Lastly, this telecoms company should focus on developing user-friendly telecom services targeted at non-“techie” customers to enhance their experience and increase retention rates.

It is important to note that correlation does not imply causation. Therefore, it is imperative to continue monitoring churn rates to assess the effectiveness of these recommendations.

5.3 Further Analysis Ideas

During the analysis, several features in the data caught my attention for future exploration. Although not directly applicable to this project, they hold potential for further investigation. One idea for future analysis is to understand the reasons for the low customer representation between 24 and 36 months ago and explore the factors contributing to this gap in tenure from the customer's perspective. Additionally, investigating whether the factors correlated with high churn rates differ between newer and long-term customers would provide valuable

insights. Furthermore, examining the variables that contribute to better retention rates in newer and long-term customers who are at risk for high churn rates could uncover important retention strategies tailored to each group.

Finally, this telecoms company should explore the relationship between streaming services and their impact on retention rates for both "at risk" and non-"at risk" groups. Unfortunately, the dataset lacks explicit information regarding the telecoms company's affiliation with streaming services. While numerous telecom companies provide bundled packages incorporating streaming services, the dataset does not clarify whether such packages have impacted the data under analysis. It is possible the inclusion of these streaming services in bundles adversely affects customer retention rates within the "at-risk" customer segments. Additionally, it is conceivable that customers who already subscribe to streaming services independent of any potential bundles offered by this telecoms company exhibit higher churn rates. Consequently, gathering further data is imperative to obtain valuable insights and facilitate comprehensive analyses in the future.